Respiratory Tract Infections

CHAPTER 4.8	RESPIRATORY TRACT INFECTIONS		
Infections and Likely Organisms	**Suggested Treatment**		**Notes**
	Recommended	**Alternative**	
Community-Acquired Bacterial Pneumonia, Pleural Effusion and Empyema			
1 - 3 months			
Streptococcus pneumoniae *Haemophilus influenzae* *Staphylococcus aureus*	Ampicillin 50 mg/kg/dose IV q6H **AND /OR** Cloxacillin 50 mg/kg/dose IV q6H	Co-amoxiclav 25-30 mg/kg/dose (Amoxicillin component) IV q8H **OR** Cefuroxime 25 mg/kg/dose IV q8H	• Use Cloxacillin if one suspects *Staphylococcus aureus* (MSSA) pneumonia / sepsis in critically ill infants, children < 1 year of age or in the presence of pneumatoceles. • For patients with significant beta-lactam allergy, macrolides are alternatives. • Add IV Metronidazole or change Amoxicillin to Co-amoxiclav if aspiration of anaerobes is a possibility. • Use Co-amoxiclav 30 mg/kg/dose for pneumonia with acute otitis media.
Chlamydia trachomatis *Bordetella pertussis*	Erythromycin 15 mg/kg/dose PO q6H **OR** Azithromycin **(preferred choice for infant < 1 month old):**		

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	<u>< 6 months old:</u> 10 mg/kg/dose PO q24H for 5 days **<u>≥ 6 months old:</u>** 10 mg/kg/dose PO q24H on Day 1 then 5 mg/kg/dose PO q24H on Day 2 – 5		
3 months - 5 years			
Streptococcus pneumoniae *Haemophilus influenzae* *Staphylococcus aureus*	**Outpatient:** Amoxicillin 25-30 mg/kg/dose PO q8H	**Outpatient:** Co-amoxiclav 25 mg/kg/dose (Amoxicillin component) PO q12H **OR** Cefuroxime 15 mg/kg/dose PO q12H	• Use Cloxacillin if one suspects *Staphylococcus aureus* (MSSA) pneumonia / sepsis in critically ill infants, children < 1 year of age or in the presence of pneumatoceles. • For patients with significant beta-lactam allergy, macrolides are alternatives. • Add IV Metronidazole or change Amoxicillin to Co-amoxiclav if aspiration of anaerobes is a possibility. • Use Co-amoxiclav 30 mg/kg/dose for pneumonia with acute otitis media.
	Inpatient: Ampicillin 50 mg/kg/dose IV q6H	**Inpatient:**	

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		Co-amoxiclav 25-30 mg/kg/dose (Amoxicillin component) IV q8H **OR** Cefuroxime 25-30 mg/kg/dose IV q8H	
Mycoplasma pneumoniae	Erythromycin Ethyl Succinate 20 mg/kg/ dose PO q12H	Azithromycin 10 mg/kg/dose PO q24H on Day 1 then 5 mg/kg/ dose PO q24H on Day 2 – 5	
>5 years			
Streptococcus pneumoniae	**Outpatient:** Amoxicillin 15 mg/kg/ dose PO q8H	**Outpatient:** Co-amoxiclav 25 mg/kg/ dose (Amoxicillin component) PO q12H	• Use Cloxacillin if one suspects *Staphylococcus aureus* (MSSA) pneumonia / sepsis in critically ill infants, children < 1 year of age or in the presence of pneumatoceles. • For patients with significant beta-lactam allergy, macrolides are alternatives. • Add IV Metronidazole or change Amoxicillin to Co-amoxiclav if aspiration of anaerobes is a possibility. • Use Co-amoxiclav 30 mg/kg/dose for pneumonia with acute otitis media.
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		Co-amoxiclav 25-30 mg/kg/dose (Amoxicillin component) IV q8H **OR** Cefuroxime 25-30 mg/kg/dose IV q8H	
Mycoplasma pneumoniae	Erythromycin Ethyl Succinate 20 mg/kg/dose PO q12H		
Moraxella catarrhalis	Co-amoxiclav 25 mg/kg/dose (Amoxicillin component) PO q12H		
Non-cystic Fibrosis Bronchiectasis			
Patient with no previous bacteriology result.	Co-amoxiclav 25-30 mg/kg/dose (Amoxicillin component) IV q8H	Cefuroxime 25-30 mg/kg/dose IV q8H **OR** Ceftriaxone 50 mg/kg/dose IV q24H	

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<tr><td>In patients chronically colonized with Pseudomonas aeruginosa
2 consecutive positive cultures or 3 positive cultures in 1 year</td><td>Consider adding Nebulised Amikacin for duration of 1 to 3 months*.</td><td></td><td>• Follow the most recent sputum bacteriology result in deciding which antibiotics to use.
• The total duration of antibiotics should be for 14 days.
• Azithromycin 5 mg/kg/dose (Monday, Wednesday, Friday) is considered especially when there are frequent exacerbations and decline in lung function.
• *Nebulised Amikacin dose:
<table><tr><th>Age</th><th>Dose</th></tr><tr><td>< 1 year</td><td>62.5 mg q12H</td></tr><tr><td>1 - 5 years</td><td>125 mg q12H</td></tr><tr><td>6 - 12 years</td><td>250 mg q12H</td></tr><tr><td>> 12 years</td><td>500 mg q12H</td></tr></table></td></tr>
<tr><td colspan="4">Cystic Fibrosis Antibiotic Prophylaxis</td></tr>
<tr><td colspan="4">Please refer to Chapter 3.18: Non-surgical Prophylaxis.</td></tr>
<tr><td colspan="4">Infective Pulmonary Cystic Fibrosis Exacerbation</td></tr>
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Methicillin-sensitive *Staphylococcus aureus* (MSSA)	**Outpatient:** **If patient is on Cloxacillin prophylaxis** Co-amoxiclav 25 mg/kg/dose (Amoxicillin component) PO q12H	**Outpatient:** **If patient is on Cloxacillin prophylaxis** Co-trimoxazole (Trimethoprim component) 5mg/kg/dose PO q12H **OR** Doxycycline: **<u>8-11 years:</u>** 4.4 mg/kg/dose PO q24H on Day 1 then 2.2 mg/kg/dose (can increase up to 4.4 mg/kg/dose) PO q24H thereafter for 2-4 weeks **<u>>12 years:</u>** 200 mg q24H on Day 1 then 100 mg q24H thereafter (can increase to 200 mg) for 2-4 weeks	• If normal liver function, can consider combination of: Rifampicin 10 mg/kg/dose PO q12H **PLUS** Fusidic acid 10 mg/kg/dose PO q8H
	Inpatient:		

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	Cloxacillin 50 mg/kg/ dose IV q6H		
Methicillin-resistant *Staphylococcus aureus* (MRSA)	**Outpatient:** Rifampicin 10 mg/kg/ dose PO q12H + Co-trimoxazole (Trimethoprim component) 5 mg/kg/dose PO q12H for 2 weeks **OR** Clindamycin 10 mg/kg/ dose PO q8H*	**Outpatient:** Rifampicin 10 mg/kg/ dose PO q12H + Sodium Fusidate 10-15 mg/kg (250 - 500 mg) PO q8H for 2 weeks **Resistance/treatment failure:** Linezolid used in isolation for 2 weeks. **< 12 years:** 10 mg/kg/dose q8H **> 12 years:** 600 mg q12H	• Clindamycin is used for community-acquired MRSA. • Surface swabs are taken for surveillance which includes right and left nostril, inguinal and sputum. • For MRSA nasal carriage: Topical Mupirocin to anterior nares. Patients will have to use Chlorhexidine wash on the skin once daily for 5 days. • Following treatment, sputum samples are obtained and repeated bi-monthly. If positive on the 2nd screen, oral treatment is repeated. If remains positive following 2 full courses of oral treatment or during an infective exacerbation, then consider admission and IV Vancomycin for 14 days.
	Inpatient:		

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	Vancomycin 15 mg/kg/ dose IV q8H		
Streptococcus pneumoniae *Haemophilus Influenzae*	**Outpatient:** Co-amoxiclav 25 mg/kg/ dose (Amoxicillin component) PO q12H for 4 weeks	**Outpatient:** Cefuroxime 15 mg/kg/ dose PO q12H for 2 weeks	
	Inpatient: Co-amoxiclav 25-30 mg/kg/dose (Amoxicillin component) IV q8H for 2 weeks	**Inpatient:** Cefuroxime 30 mg/kg/ dose IV q8H for 2 weeks	
Pseudomonas aeruginosa	**First isolate:** Ciprofloxacin 20 mg/kg/ dose PO q12H (max dose 750 mg q12H) + Nebulised Gentamicin or Amikacin for 3 months		• **Precaution:** Use of quinolones in paediatric patients is associated with increased incidence of adverse reaction related to joint i.e., tendonitis, tendon rupture • If the repeat sputum is negative for PA, then stop the oral Ciprofloxacin and continue nebulised Amikacin or Gentamicin for a total of 3 months.

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	Re-culture the sputum after 3 weeks. If positive: Ceftazidime 50 mg/kg/ dose IV q8H (max 9 g/ day) + Amikacin 30 mg/kg/dose IV q24H (max 2 g) for 2 weeks		• If there is a substantial Pseudomonal free period (> 3 months), the patient is treated in same manner as first isolate. • To monitor serum drug concentration of Amikacin.
Chronic *Pseudomonas* Infection *Defined as 3 positive cultures at least 2 months apart and within 1 year*.	**During exacerbation,** **Outpatient:** Ciprofloxacin 20 mg/kg/ dose PO q12H (max dose 750 mg q12H) for 2 weeks		• Once chronic infection is established, there is a trend for the *Pseudomonas* to become mucoid and more resistant to antibiotics. The aim of further treatment is to reduce the colony numbers in order to suppress the inflammatory response. • For chronically colonized patient, alternating monthly treatment of nebulised antibiotics (Gentamicin / Amikacin/ Colistin) for 3 months i.e., Neb Gentamicin/ Amikacin/ Gentamicin or Neb Amikacin/ Colistin/ Amikacin.
Chronic *Pseudomonas* Infection	**Inpatient or elective tuning:** Ceftazidime 50 mg/kg/	**Inpatient or elective tuning:**	• Once chronic infection is established, there is a trend for the *Pseudomonas* to become mucoid and more resistant to

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Defined as 3 positive cultures at least 2 months apart and within 1 year.	dose IV q8H (max 9 g/ day) + Amikacin 30 mg/kg/dose IV q24H (max 2 g/dose) **OR** Gentamicin 10 mg/kg/ dose IV q24H for 2 weeks **Followed by nebulised Amikacin or Gentamicin for 3 months**	Piperacillin/Tazobactam 100 mg/kg/dose (Piperacillin component) IV q8H **OR** Meropenem 40 mg/kg/ dose IV q8H + Amikacin 30 mg/kg/dose IV q24H (max 2 g/dose) **OR** Gentamicin 10 mg/kg/ dose IV q24H for 2 weeks **For multidrug-resistant *Pseudomonas* spp:** Polymyxin B: **< 2 years old:** 0.75 - 2 mg/kg/dose IV q12H over 60-90 min **> 2 years old:** 0.75 - 1.5 mg/kg/dose IV q12H	antibiotics. The aim of further treatment is to reduce the colony numbers in order to suppress the inflammatory response. • For chronically colonized patient, alternating monthly treatment of nebulised antibiotics (Gentamicin / Amikacin/ Colistin) for 3 months i.e., Neb Gentamicin/Amikacin/ Gentamicin or Neb Amikacin/ Colistin/Amikacin. • To monitor serum drug concentration of Amikacin.

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<tr><td></td><td></td><td>over 60-90 min</td><td></td></tr>
<tr><td>Burkholderia cepacia complex</td><td>Ceftazidime 50 mg/kg/dose IV q8H (max 9 g/day)
+
Meropenem 40 mg/kg/dose IV q8H (max 2 g q8H) for 2 weeks</td><td>Chronic Burkholderia cepacia complex:
Doxycycline:
8-11 years:
4.4 mg/kg/dose PO q24H on Day 1 then 2.2 mg/kg/dose (can increase up to 4.4 mg/kg/dose) PO q24H thereafter for 2-4 weeks
>12 years:
200 mg q24H on Day 1 then 100 mg q24H thereafter (can increase to 200 mg) for 2-4 weeks.

Nebulised Meropenem*</td><td>*Nebulised Meropenem:<table><tr><th>Age</th><th>Dose</th></tr><tr><td>6-12 years</td><td>125 mg q12H</td></tr><tr><td>12 years</td><td>250 mg q12H</td></tr></table></td></tr>
<tr><td>Stenotrophomonas maltophilia</td><td>Outpatient:</td><td>Outpatient:</td><td></td></tr>
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	Co-trimoxazole (Trimethoprim component) 5 mg/kg/ dose PO q6-12H for 4 weeks.	Levofloxacin 8-10 mg/kg/dose PO q12H **OR** Doxycycline: **8-11 years:** 4.4 mg/kg/dose q24H on day 1 then 2.2 mg/kg/dose (can increase up to 4.4 mg/kg/dose) q24H thereafter for 2-4 weeks **>12 years:** 200 mg q24H on day 1 then 100 mg q24H thereafter (can increase to 200mg) for 2 -4 weeks	
	Inpatient: Co-trimoxazole (Trimethoprim component) 5 mg/kg/ dose IV q6-12H	**Inpatient:** Levofloxacin 8-10 mg/kg/dose IV q12H **OR** Ceftazidime 50 mg/kg/ dose IV q8H	

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Allergic Bronchopulmonary *Aspergillosis*	**Prednisolone** 2 mg/kg/day for 2 weeks, reduced to 1 mg/kg/day for 2 weeks, then alternate 1 mg/kg/day for 2 weeks. Wean slowly over the next 4 months, reducing dose by 2.5 mg or 5 mg every 2 weeks. **PLUS** **Itraconazole** **1 month - 12 years:** 5 mg/kg/dose PO q12h (max 200 mg q12H) **> 12 years:** 200 mg q12H (stopped 2-4 weeks after the oral steroids are finished) **Nebulised**	**Pulsed Methylprednisolone** 10 - 15 mg/kg/dose (max 1 g/dose) IV q24H for 3 days every month. **Voriconazole** as a second-line agent for patients who have not responded to or cannot tolerate twice daily Itraconazole: **2-11 years:** 9 mg/kg/dose (max 350 mg) q12H **12-14 years:** • **< 50 kg:** 9 mg/kg/dose (max 350 mg) q12H • **> 50 kg:** 400 mg q12H for 2 doses then 200 mg q12H (max 300 mg q12H) **15 years and above:** • **< 40 kg:** 200 q12H for 2 doses then 100 mg	• The response to treatment is monitored by checking pulmonary function tests, CXRs and watching IgE titre as this should fall over time to more normal levels. • Monitor liver function.

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	Amphotericin B (non-liposomal) may be used in difficult cases after physiotherapy: **< 10 years:** 5 mg q12H **> 10 years:** 10 mg q12H	q12H (max 150 mg q12H) • **> 40 kg:** 400 mg q12H for 2 doses then 200 mg q12H (max 300 mg q12H)	
Invasive *Aspergillus*	Amphotericin B Deoxycholate 0.5 -1 mg/kg/dose IV q24H	Voriconazole 9 mg/kg/ dose IV q12H for 2 doses, followed by 8 mg/kg/dose IV q12H	